Paleomagnetism and K-Ar Ages of Some Trachybasaltic Dikes From Río de Los Molinos, Province of Córdoba, Republic of Argentina

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Measurements of the K-Ar ages of eight trachybasaltic dikes from the Río de Los Molinos zone, Argentina, suggest that the dikes were intruded during at least two magmatic episodes at about 140 and 65 m.y. B.P. Normal and reversed magnetic remanences are observed in the older dikes; these suggest that during the older magmatic episode at least two events of intrusion occurred; a minimum period of 5000 years must have elapsed between these intrusions. The pole computed for the Upper Jurassic–Lower Cretaceous dikes (79°S, 8°E, α_{95} = 18°) is reasonably close to the paleomagnetic poles of the Vulcanitas Cerro Colorado formation (118 m.y.) and the Serra Geral Formation (115–125 m.y.). This supports the interpretation that the South Atlantic Ocean was opened in Lower Cretaceous times.

INTRODUCTION

Poles computed for the igneous rocks of the Serra Geral [*Creer*, 1962] and the Vulcanitas Cerro Colorado formations [*Valencio*, 1972] are the only paleomagnetic poles of Lower Cretaceous age available for South America. Paleomagnetic poles for the Cretaceous La Teta lava [*McDonald and Opdyke*, 1972] and the Late Cretaceous Poços de Caldas alkaline complex [*Opdyke and McDonald*, 1973] have also been reported.

Paleomagnetic reconstructions of the ancient relative positions of continents should be based only on paleomagnetic poles. The La Teta lava appears to have been rotated tectonically [*McDonald and Opdyke*, 1972], so that a direct comparison of its pole with the poles of South America and Africa is of little value in evaluating the relative positions of these continents in Cretaceous times.

Comparison of the South American paleomagnetic poles with those of the same geological age for Africa has suggested that the fragmentation of the South American–African block and the origin of the South Atlantic Ocean occurred before the Lower Cretaceous [*Creer*, 1962; *Valencio*, 1972] but after the Middle Jurassic [*Creer et al.*, 1972]. More Cretaceous paleomagnetic poles for South America are necessary to support this interpretation. Evidence for the South Atlantic from magnetic anomaly studies [*Heirtzler et al.*, 1968; *Ladd et al.*, 1975] and from the Deep-Sea Drilling Project [*Maxwell et al.*, 1970] suggests that the South Atlantic had opened to about one half of its present width by the Cretaceous times.

Trachybasaltic dikes crop out at Río de Los Molinos, in the southern part of the Sierra Chica geologic unit of Córdoba province, Republic of Argentina. *Gordillo and Lencinas* [1969] have suggested a Lower Cretaceous age for these trachybasaltic dikes. However, a younger K-Ar age (60 ± 3 m.y.) has been obtained for one of the dikes of Río de Los Molinos. This age is questioned on the grounds that it may have been reset [*Gordillo and Lencinas*, 1969].

The authors have carried out a radiometric and paleomagnetic study of the trachybasaltic dikes from Río de Los Molinos. In this paper the K-Ar ages and the paleomagnetic data obtained for eight of these dikes are discussed.

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GEOLOGICAL EVIDENCE

The Río de Los Molinos zone is located in the west central part of the Córdoba province, Republic of Argentina, at about 64°31'W, 31°48'S (Figure 1). *Gordillo and Lencinas* [1969] have reported that the crystalline basement, probably of late Precambrian age, is fairly well exposed at the Río de Los Molinos zone. The basement is formed by units of tonalitic gneiss alternating with amphibolites and marbles. Gordillo and Lencinas report that the structure in the zone is typified by fault blocks, which are bounded by high-angle reverse faults. In Pliocene-Pleistocene times these faults were reactivated, and the blocks were elevated and gently inclined to the east 2°–4°. The last igneous event in the area was the intrusion of trachybasaltic dikes. *Gordillo and Lencinas* [1969] have suggested that the dikes are chemically and mineralogically related to the Lower Cretaceous trachybasalts of the Sierra de Los Cóndores group, which are exposed about 35 km south of the Río de Los Molinos zone [*Gordillo and Lencinas*, 1967]. The dikes under study in the present paper have a NNW-SSE strike and a near-vertical dip. Their mean width is of the order of 0.7 m.

Gordillo and Lencinas [1969] assume that the dikes of the Río de Los Molinos zone are of the same age as the vulcanites outcropping at Sierra de Los Cóndores (118 m.y., Lower Cretaceous), but a K-Ar date for one of the dikes (14) measured by U. Cordani from the University of São Paulo, Brazil, indicates an age of 60 ± 3 m.y. (lower Cenozoic).

POTASSIUM-ARGON AGES

Many of the trachybasaltic dikes of the Río de Los Molinos zone are badly weathered, and it was possible to obtain good samples from only eight of them. Sampling was carried out along route 36, on the basis of the geological study done by *Gordillo and Lencinas* [1969] (Figure 1). Twenty-six oriented

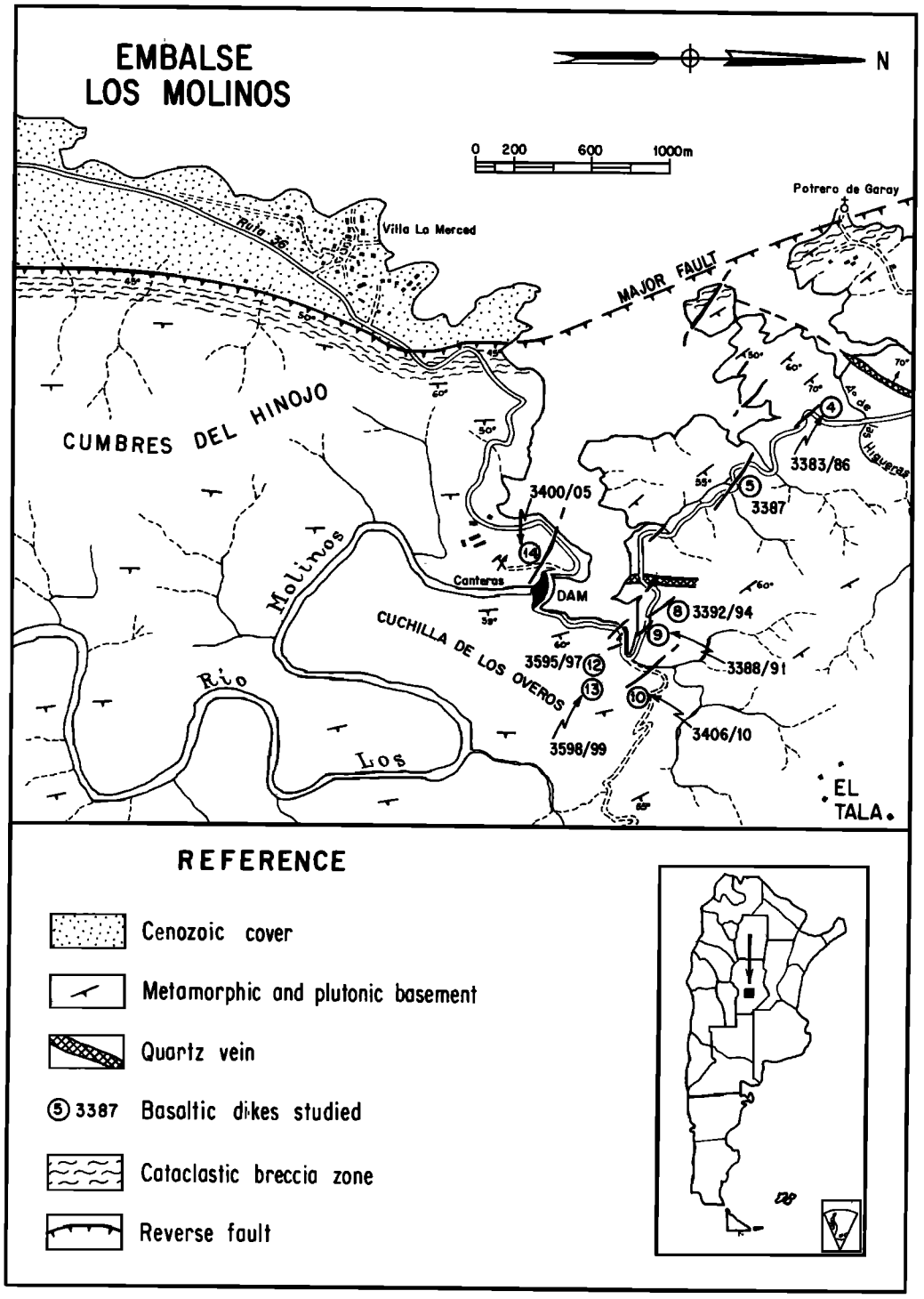


Fig. 1. Schematic geological map of the Río de Los Molinos zone, province of Córdoba, Argentina, according to *Gordillo and Lencinas* [1969]. The trachybasaltic dikes are numbered according to these authors. The identification numbers of the hand samples collected from each dike are also shown.

hand samples were collected. Samples showing the least alteration were dated by the K-Ar method at the Instituto de Geocronología y Geología Isotópica, University of Buenos Aires, Argentina.

The argon extraction was accomplished by the isotope dilution method, and two determinations were made by using two different fractions of each sample. The potassium analysis was performed in duplicate for each sample. The equipment and techniques used in both determinations are described by *Linares et al.* [1973], who showed that the standard deviations are of the order of 2.5–3% for the argon determination and about 1% or less for the potassium analyses.

The age and analytical error for each sample were calculated according to the method of *Cox and Dalrymple* [1967]. The argon extraction for each sample was measured twice at different places, and the best/fitting $^{40}Ar/^{38}Ar$ and $^{38}Ar/^{38}Ar$ ratios were determined by the least squares method. In this way we obtained two ages for each sample analyzed, and in all

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TABLE 1. Analytical Data of the Radiometric Age Determinations for the Trachybasaltic Dikes From Río de Los Molinos, Córdoba Province, Argentina

Dike	Sample Number	K, %	$^{40}Ar_{rad}$, 10^{-10} mol/g	Average $^{40}Ar_{rad}$, 10^{-10}mol/g	$^{40}Ar_{atm}$, %	Age, m.y.	Average Age, m.y.
8	3392/94	1.16	3.286	3.185	47.5	149	150 ± 10
			3.083		45.3	151	
5	3387	2.08	5.180	5.392	34.2	136	140 ± 10
			5.604		27.9	143	
10	3406/10	0.59	1.546	1.503	53.8	142	138 ± 10
			1.459		83.1	134	
9	3389/91	0.92	2.267	2.218	16.0	134	131 ± 10
			2.169		47.3	128	
12	3395/96	1.63	3.726	3.656	43.0	124	129 ± 10
	3397		3.587		62.5	122	
		1.34	3.316	3.316	55.3	135	
4	3383/86	1.18	1.494	1.448	36.0	70	68 ± 5
			1.401		60.0	66	
13	3398/99	0.63	0.686	0.737	92.5	62	65 ± 5
			0.788		73.4	68	
14	3401	1.61	2.065	1.957	21.0	70	63 ± 5
			1.849		23.9	64	
14	3405	1.88	1.807	1.948	24.3	62	63 ± 5
			1.932		24.7	56	
			2.024		81.0	57	

cases the differences were small. Table 1 shows the analytical data of the radiometric age determinations for the eight dikes involved in this work.

Two of the analyzed samples came from the same dike (14) previously dated at 60 ± 3 m.y. in the São Paulo laboratory. The five ages obtained for these samples range between 56 and 70 m.y., the average age being 63 ± 5 m.y. (Table 1).

The K-Ar ages for the trachybasaltic dikes suggest that in the Río de Los Molinos zone there were at least two periods of intrusion: dikes 5, 8, 9, 10, and 12 were formed in the range 129–150 m.y., and dikes 4, 13, and 14 were intruded during a younger magmatic event at about 65 m.y. The K-Ar isochron method [*Hayatsu and Carmichael*, 1970] was also applied to the data. Figure 2 shows an isochron plot of the $^{40}Ar/^{36}Ar$

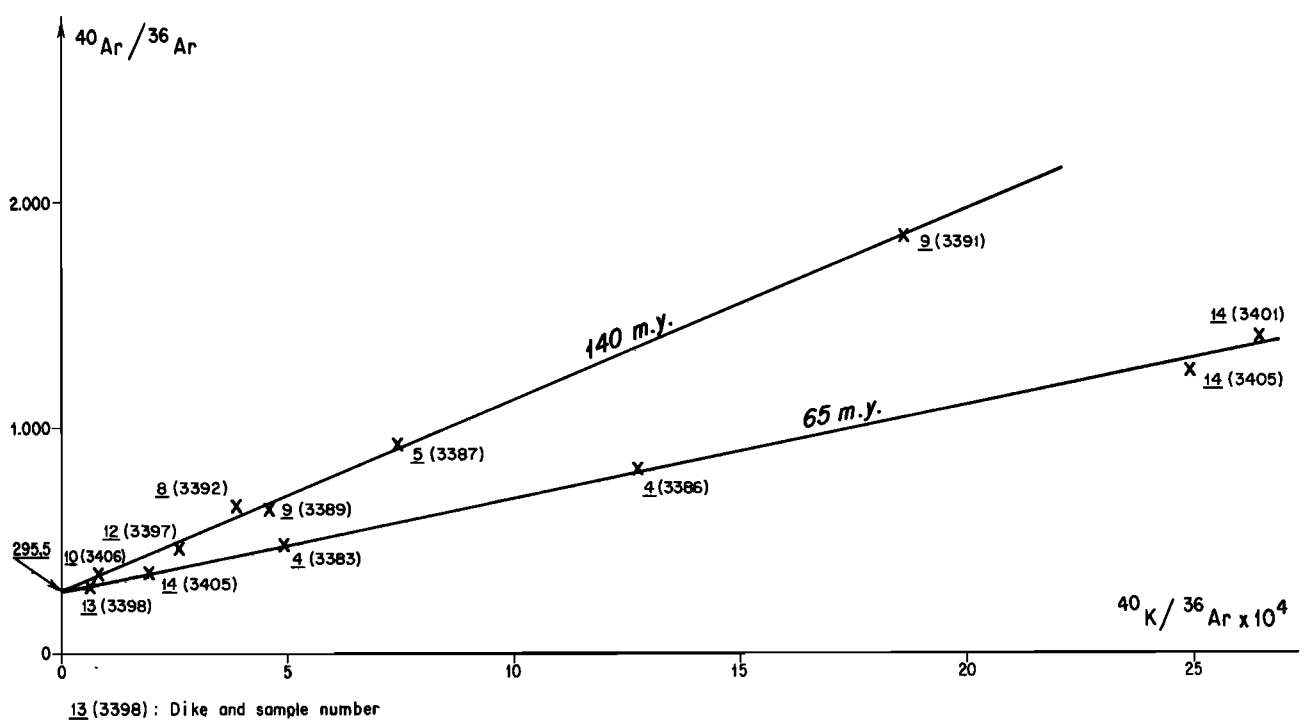


Fig. 2. Isochron plotting of the samples from the Río de Los Molinos zone.

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TABLE 2. Summary of the Paleomagnetic Data for the Trachybasaltic Dikes From the Río de Los Molinos Zone, Sierra Chica, Córdoba Province, Argentina

		Direction of Cleaned Remanent Magnetization						Paleomagnetic Pole						
Dike	Sample Number	N	D, deg	I, deg	R	α_{95}, deg	Polarity*	Lat., °S	Long., °E	$d\psi$, deg	$d\chi$, deg	α_{95}, deg	Pole Name	K-Ar Age, m.y.
8	3392–94	3	163	55	2.96	17	R	75	7	17	24		8	150 ± 10
5	3387	1	346	−46			N							140 ± 10
10	3406–07–10	2	187	55	1.98	34	R	82	244	35	48		10	138 ± 10
9	3388–91	2	149	40	1.98	37	R	61	36	26	44		9	131 ± 10
12	3395–97	3	352	−58	2.97	15	N	80	332	17	22		12	129 ± 10
4	3383–86	4	164	55	3.97	8	R	76	24	12	19		4	68 ± 5
13	3398–99	2	168	49	1.99	19	R	79	30	17	26		13	65 ± 5
14	3400–05	6	162	55	5.81	13	R	75	4	13	19		14	63 ± 5
Older dikes			166	51	4.92	11	N-R	79	8			18		140
Younger dikes			165	53	2.96	6	R	77	18			5		65
All dikes			165	52	7.91	6	N-R	78	13			8	SAK_3	

N, number of unit vectors; D, declination; I, inclination; R, length of the resultant vector; $d\psi$, semiminor axis oval of confidence in which the true pole position lies with 95% probability; $d\chi$, semimajor axis oval of confidence in which the true pole position lies with 95% probability.
* N indicates normal, and R indicates reversed.

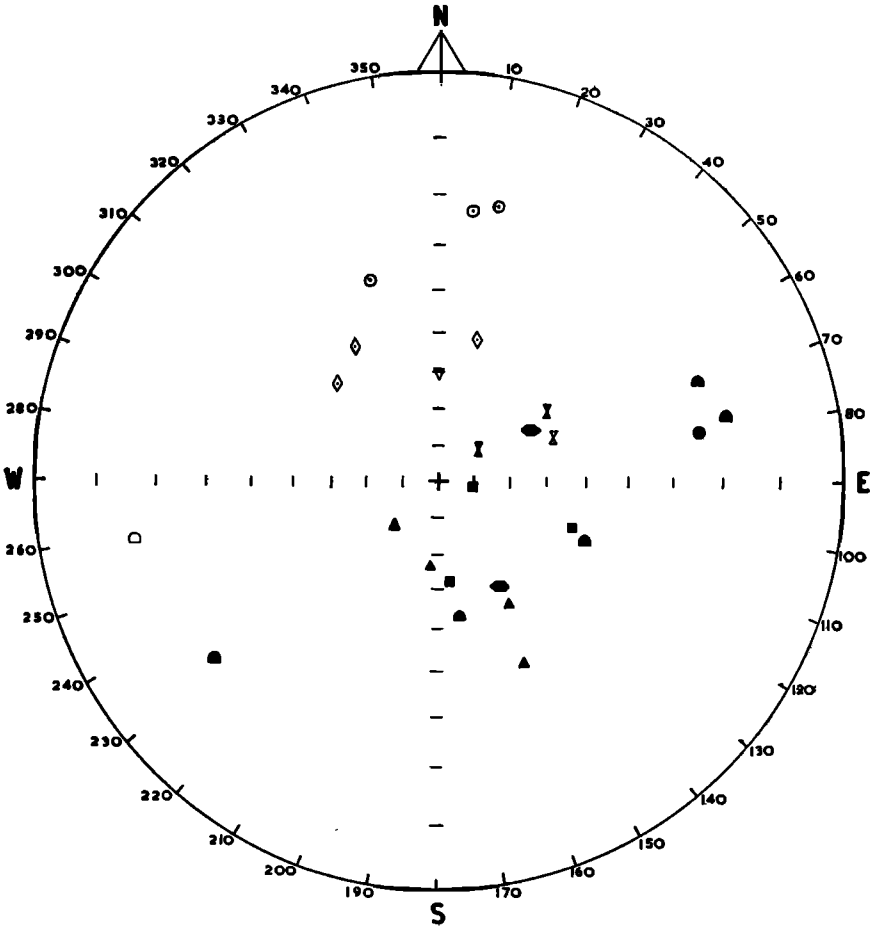


Fig. 3*a*

Fig. 3. Directions of (*a*) NRM and (*b*) cleaned remanence of hand samples. The symbols used are as follows: dike 4, circle; dike 5, inverted triangle; dike 8, square; dike 9, triangle; dike 10, hourglass; dike 12, diamond; dike 13, hexagon; and dike 14, half circle. Open symbols indicate normal cleaned remanent magnetization, and solid symbols indicate reversed cleaned remanence. The direction of the present geomagnetic field is also indicated (plus sign).

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ratio versus the $^{40}K/^{36}Ar$ ratio for each sample. The data for dikes 5, 8, 9, 10, and 12 lie along one line, whereas the data from dikes 4, 13, and 14 fall along a different line, suggesting that resetting of the ages is improbable and supporting the contention that two different magmatic episodes occurred. The mean age for the older dikes is 140 m.y., whereas the mean age for the younger dikes is 65 m.y. Therefore the radiometric study suggests that in Río de Los Molinos at least two periods of intrusion occurred.

In both cases shown in Figure 2 the $^{40}Ar/^{36}Ar$ initial ratios correspond to the accepted value of 295.5 used for the correction of the presence of atmospheric ^{40}Ar [*Nier*, 1950].

THE PALEOMAGNETIC STUDY

The orientation of the 26 hand samples collected from the trachybasaltic dikes of the Río de Los Molinos zone was carried out with both a magnetic and a sun compass.

Four disks 1 cm thick and 2.5 cm in diameter were cut from each hand sample. A spinner magnetometer was used to measure the natural remanent magnetization (NRM) of these disks. Detailed ac cleaning was used to determine the stability of the NRM of the samples. One disk from each hand sample was subjected to ac demagnetizing fields increasing in steps of 50 Oe up to 300 Oe. In this way the best ac demagnetizing field isolating the stable remanent magnetization of each sample was determined. The criterion adopted for the choice of the best ac demagnetizing field strength was that the direction of the remanence of the pilot disk exhibited no further change in response to higher demagnetizing field strengths. The best ac demagnetizing field strength was 150 Oe for the majority of samples. The other three disks of the same sample were subjected to this ac demagnetizing field for a better definition of the cleaned remanent magnetization of the sample.

A wide spectrum of remanent coercivities is represented in the samples collected. Samples from dikes 8 and 13 have the highest remanent coercivities. We isolate stable remanence in only two samples from dikes 9 and 10. In particular, samples from dike 14 are substantially better grouped after ac demagnetization than before it. These samples have reversed stable magnetization (Table 2); therefore the high scattering of the directions of NRM of samples from dike 14 could be the result of secondary components directed in the direction of the normal polarity field. The ac demagnetization indicates that most of the soft components of the remanence of these samples were destroyed in the first steps of the cleaning. Therefore the secondary components of the samples from dike 14 may be due to viscous remanent magnetization.

Fisher's [1953] statistics were used to compute the mean directions of (1) the remanent magnetization of each sample on the basis of the direction of the NRM (Figure 3*a*) and the cleaned remanence of its disks (Figure 3*b*) and (2) of each trachybasaltic dike (Table 2) on the basis of the mean cleaned magnetic remanence of its samples. Directions of the stable remanence of samples from each dike are substantially better grouped after ac demagnetization than before it (Figures 3*a*, *b*). The occurrence of well-grouped directions, divergent from the present geomagnetic field, and the presence of a reversal (Figure 3) mean that substantial secondary remanences are absent.

Dikes 4, 8, 9, 10, 13, and 14 are reversely magnetized, and

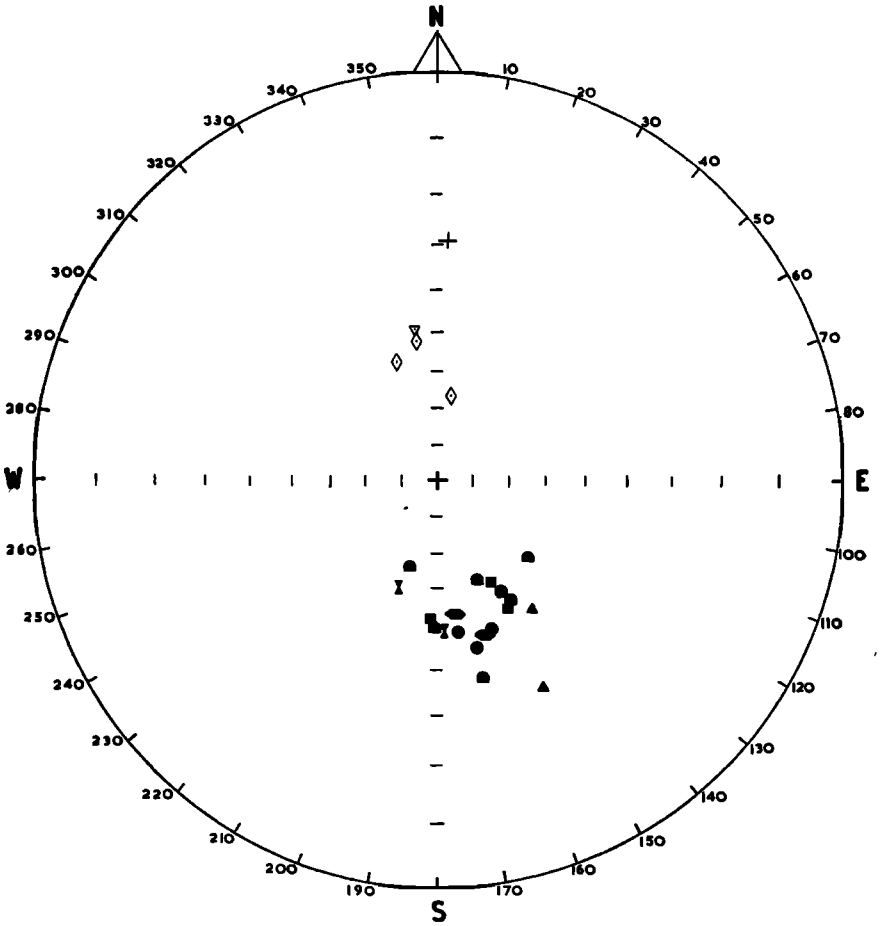


Fig. 3*b*

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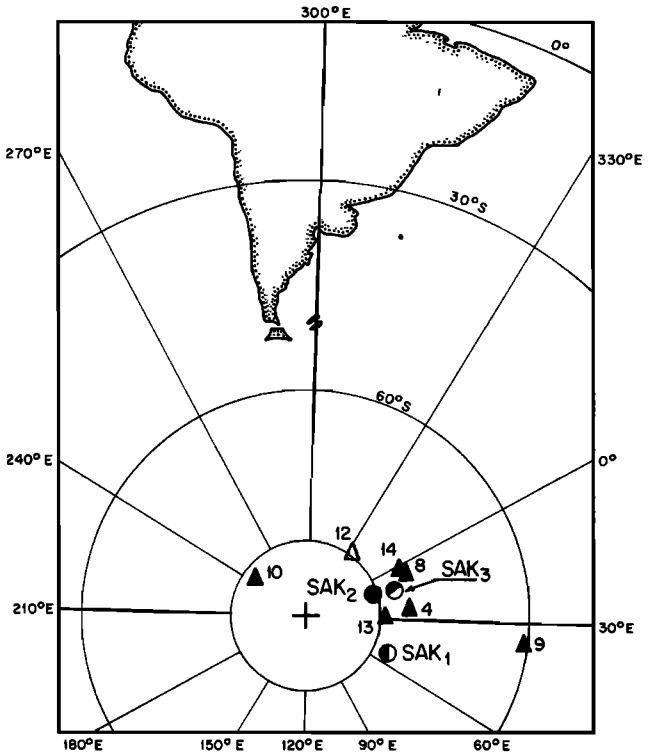


Fig. 4. Cretaceous paleomagnetic poles for South America (circles) and virtual geomagnetic poles for the trachybasaltic dikes from the Río de Los Molinos zone (triangles) numbered as in Table 1. SAK_1 and SAK_2 are the paleomagnetic poles for the Serra Geral [*Creer*, 1962] and Vulcanitas Cerro Colorado [*Valencio*, 1972] formations, respectively.

dikes 5 and 12 are normally magnetized. This suggests either that the older magmatic episode persisted during at least one change in the polarity of the geomagnetic field (presumably about 5000 years) or that during the older episode there may have been at least two magmatic events; a minimum period of 5000 years must have elapsed between these events. The number and size of the dikes and the small area covered by the dike exposures support the second interpretation.

Virtual geomagnetic poles for each of the trachybasaltic dikes are given in Table 2.

Discussion

The radiometric study has suggested that some of the trachybasaltic dikes of the Río de Los Molinos zone were intruded during two different episodes, one occurring at about 140 m.y. (Upper Jurassic–Lower Cretaceous) and the other at about 65 m.y. (lower Cenozoic). However, it has been suggested that the K-Ar ages of the younger dikes were reset in Cenozoic times [*Gordillo and Lencinas*, 1969], and we cannot rule out this possibility completely. According to present knowledge about the history of the South Atlantic Ocean, the lower Cenozoic position of South America relative to the pole was close to its position in Lower Cretaceous–Jurassic times [*Vilas*, 1970]. Therefore it is not possible to differentiate the ages of middle and upper Mesozoic and lower Cenozoic South American rocks on the basis of the direction of their remanences. Thus the paleomagnetic data do not allow us to say whether the isotopic ages of the younger dikes of the Río de Los Molinos zone were reset in lower Cenozoic times (Figure 3).

Paleomagnetic data suggest that during the older magmatic episode there were at least two events of intrusion; a minimum time span of above 5000 years must have elapsed between these events. The same interpretation is valid even if we accept that the trachybasaltic dikes were intruded during only one episode.

The large analytic errors involved in the K-Ar ages prohibit the use of the paleomagnetic and radiometric data for the trachybasaltic dikes from Río de Los Molinos to define the Mesozoic–early Cenozoic geomagnetic time scale. The Mesozoic dikes probably fall somewhere within the zone of frequent reversals of the geomagnetic field defined from the Callovian to the Barremian by Pechersky (quoted by *Creer* [1971]), which includes *McElhinny and Burek*'s [1971] Serra Geral zones.

The positions of the virtual geomagnetic poles for the trachybasaltic dikes from Río de Los Molinos (Table 2) are shown in Figure 4. The statistical analyses of these poles must be carried out on the basis of their ages. First, let us consider the interpretation suggesting that these dikes were intruded during two different episodes. The mean of the four virtual geomagnetic poles of Upper Jurassic–Lower Cretaceous age is 79°S, 8°E, α_{95} = 18°, and the mean of the three lower Cenozoic poles is 77°S, 18°E, α_{95} = 5°. The positions of these mean poles (particularly of the younger pole) may be affected by the contemporaneous nondipole components of the geomagnetic field. The second interpretation suggests that all the dikes were intruded in Upper Jurassic–Lower Cretaceous times; therefore the mean of the seven virtual geomagnetic poles is 78°S, 13°E, α_{95} = 8° (labeled SAK_3 in Figure 4).

The positions of these three poles are coincident with the level of confidence adopted and reasonably close to the positions of the paleomagnetic poles for the Lower Cretaceous Vulcanitas Cerro Colorado formation (SAK_2) and Serra Geral formation (SAK_1), which includes igneous rocks of Jurassic age, in Figure 4. This suggests that the positions of the mean of the four Upper Jurassic–Lower Cretaceous virtual geomagnetic poles and the paleomagnetic pole SAK_3 are not substantially affected by the secular variation of the geomagnetic field.

The mean of the paleomagnetic poles for the Serra Geral and Vulcanitas Cerro Colorado formations and the pole for the Upper Jurassic–Lower Cretaceous dikes from Río de Los Molinos defines the Upper Jurassic–Lower Cretaceous pole position relative to South America on the basis of the data for three different geological units (79°S, 29°E, α_{95} = 8° for SAK_3). This mean pole position differs about 26° from that of the paleomagnetic pole for the Cretaceous of Africa [*Briden*, 1967] when it is rotated with South America to Africa so as to reconstitute Gondwana (the pole of rotation is 329.4°E, 44°N, and the angle of rotation is 57° [*Smith and Hallam*, 1970]). This supports the interpretation that the South Atlantic Ocean was opened in Lower Cretaceous time.

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